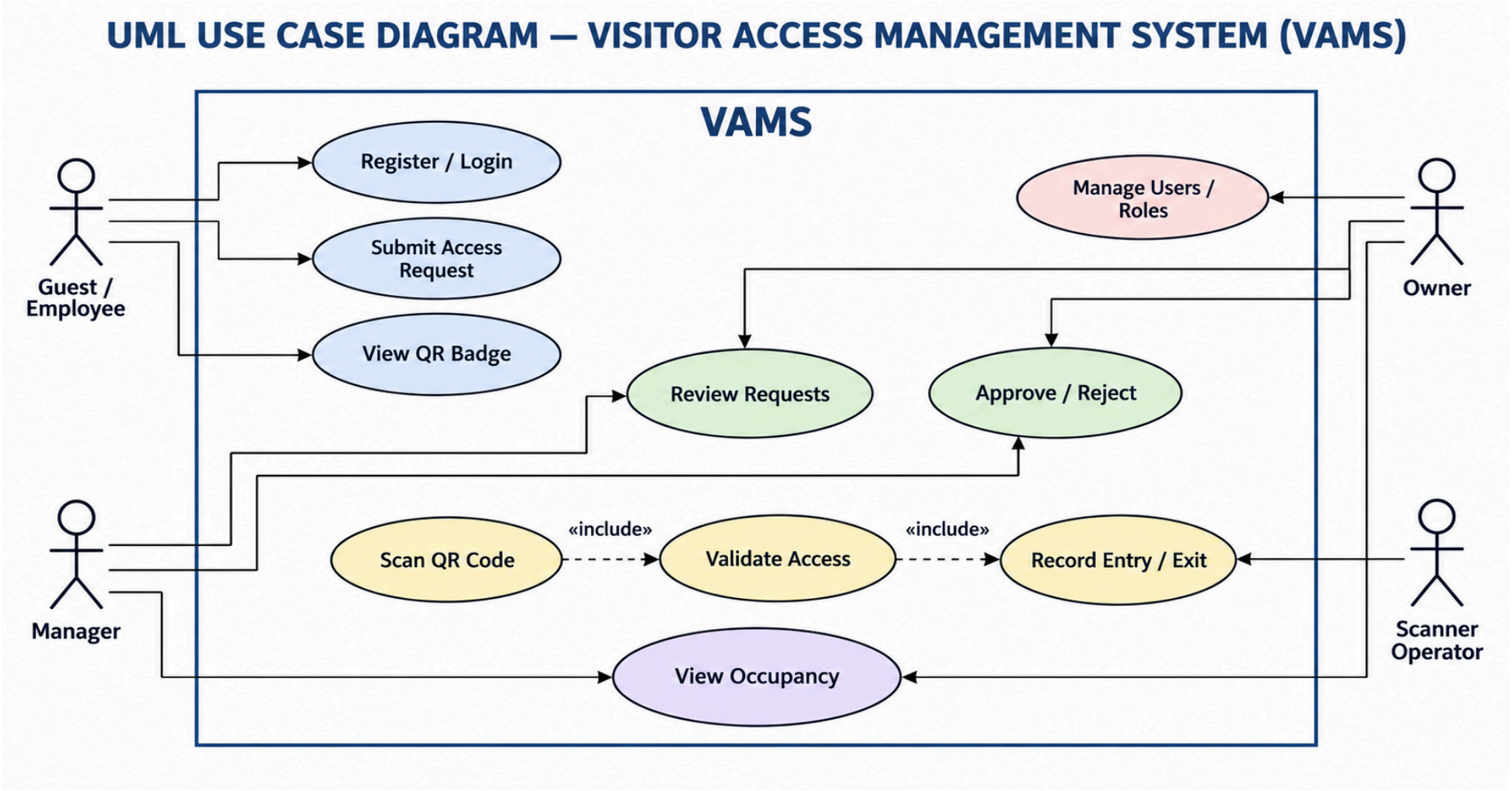
# *UML DESIGN*

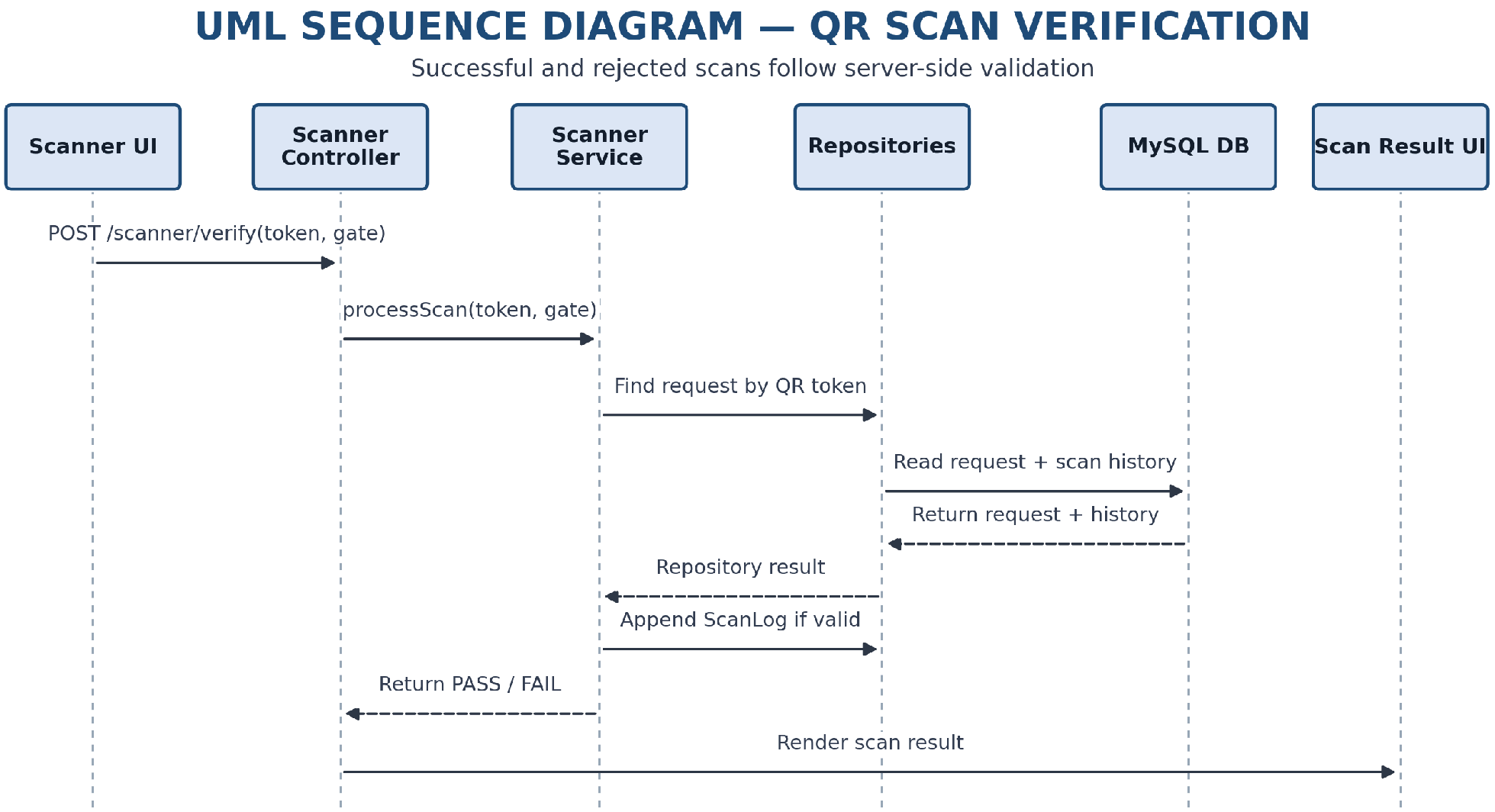
## QR-BASED VISITOR ACCESS MANAGEMENT SYSTEM (VAMS)

# 1. Use Case Diagram



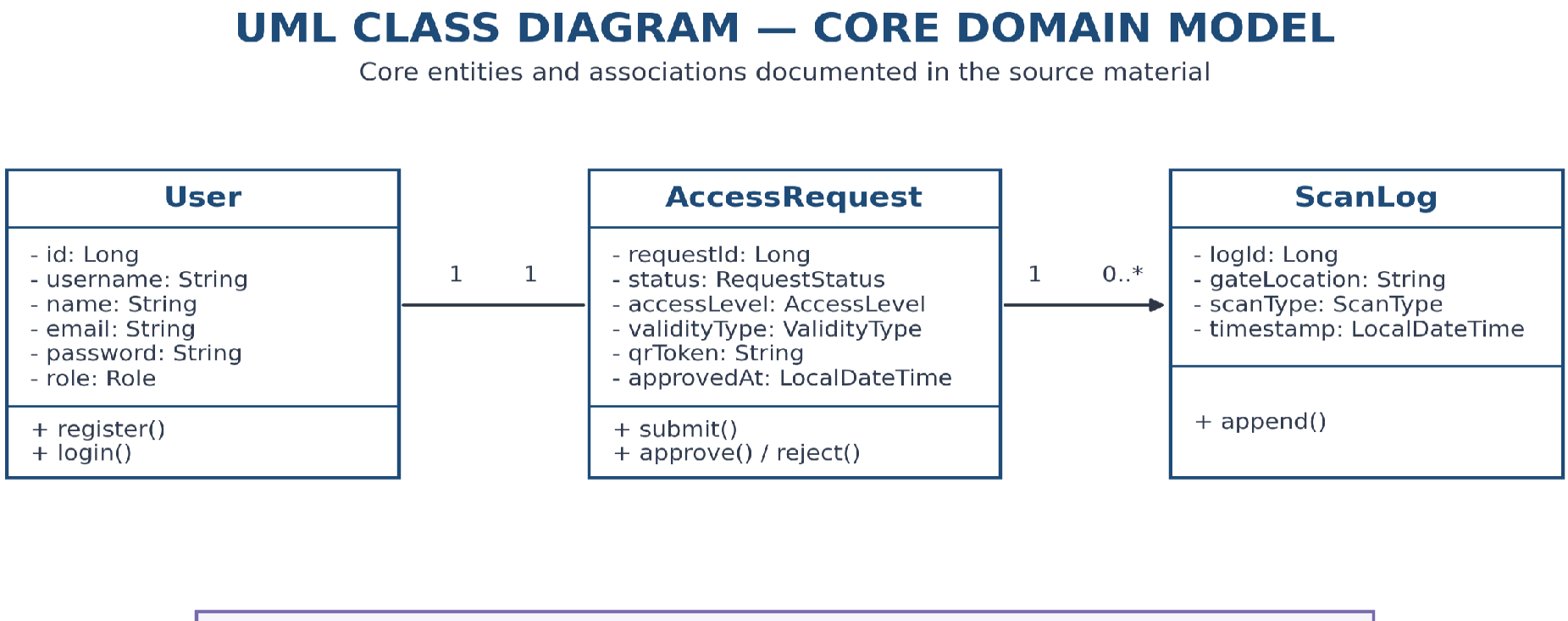

Roles: Owner, Manager, Employee, Guest, and Scanner Operator.

## 2. Sequence Diagram — QR Scan Verification




Invalid token, unapproved request, or expired ONE_DAY access returns FAIL; valid scans are appended to scan_logs.

## 3. Class Diagram — Core Domain Model

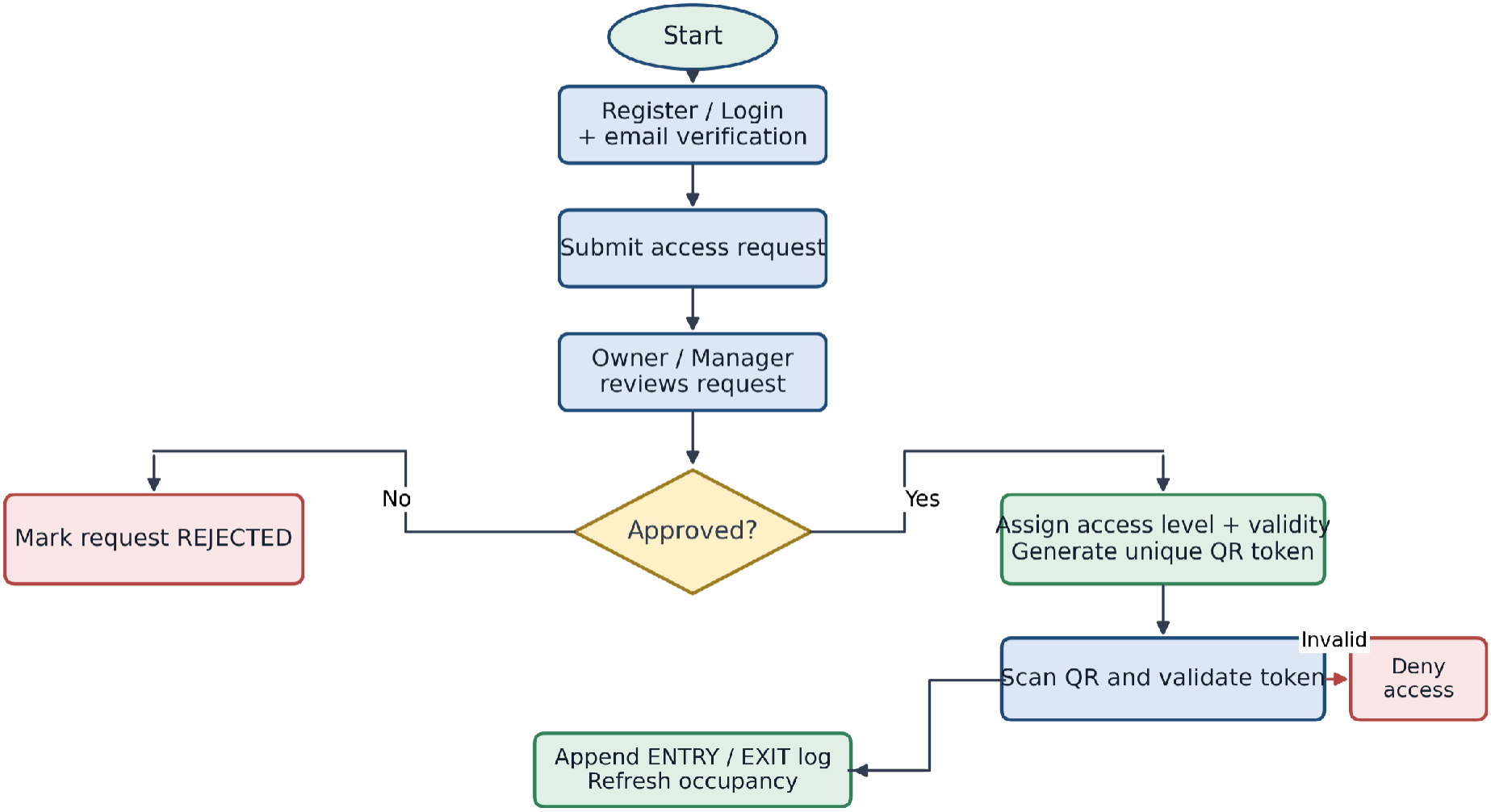



Conceptual associations; the documented core tables are users, access_requests, and scan_logs.

# 4. Activity Diagram — Registration, Approval and QR Access

## UML ACTIVITY DIAGRAM — REGISTRATION, APPROVAL AND QR ACCESS

Main user workflow, approval branch, and gate scan validation

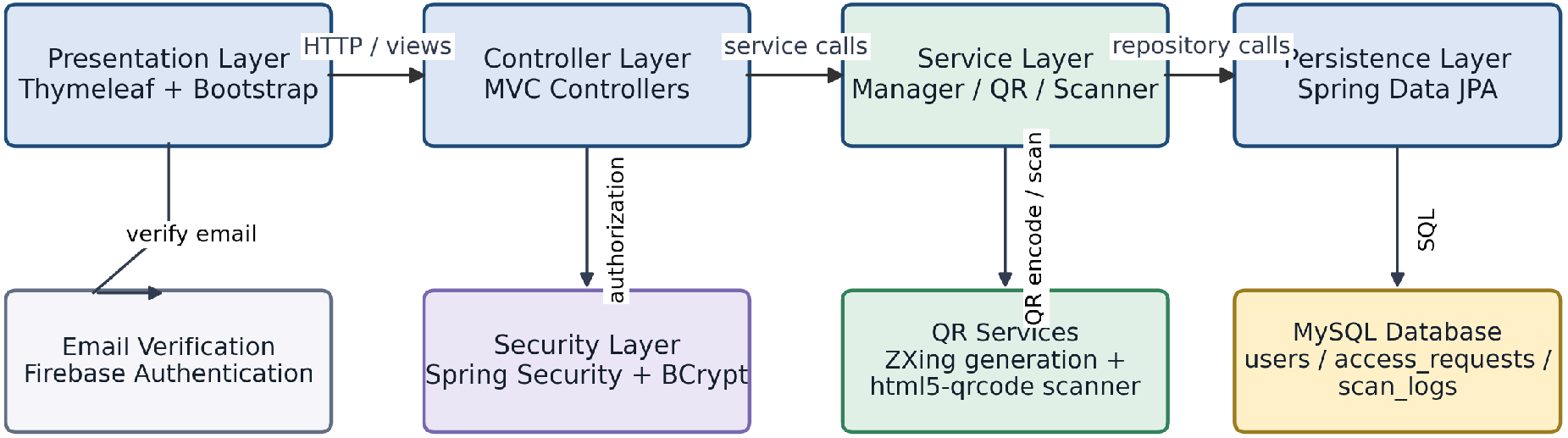



Validity types: ONE_DAY, UNTIL_LOGOFF, and ONGOING. Access level: PUBLIC or RESTRICTED.

## 5. Component Diagram — Application Architecture

# UML COMPONENT DIAGRAM — APPLICATION ARCHITECTURE

Java Spring Boot web application and documented supporting libraries




**Core data model: User • AccessRequest • ScanLog**

ScanLog is append-only; occupancy = max(0, total ENTRY − total EXIT).

The diagram reflects the documented Spring Boot, Spring Security, ZXing, html5-qrcode, Firebase, JPA, and MySQL components.